



Synchronous machine working as generator

Now, decrease the field current of the dc motor (i.e. make synchronous machine in generator mode) so that wattmeter reads a specified output (say, 600W per phase) (Please take care of the multiplication factor of wattmeter). Keeping the wattmeter reading at 600W/phase, increase the synchronous machine field current I_f by adjusting the potential divider so that ammeter reading I_a shows rated current. Note I_f and I_a . Now decrease I_f gradually and note down I_a each time. We can see that I_a decreases from rated current, reaches minimum, again increases and reaches rated value. Wattmeter reading should be same while taking each reading (Otherwise, vary motor field rheostat). I_a Vs I_f curve (V curve) is plotted for the generator for a constant power output of 1800W. Now increase I_f so that armature current is minimum. Increase the motor field current, so that wattmeter reads zero.

Synchronous machine working as motor

For getting V curves of synchronous motor on no load, switch off the supply to the dc machine and connect the dc machine to the lamp load by using DPDT switch when the wattmeter reading is zero and ammeter reading minimum. Now the synchronous machine will act as motor and dc machine as generator. Note down the wattmeter reading which shows the no-load losses of both dc machine and synchronous motor. Increase the field current I_f of synchronous machine till the armature current I_a becomes rated value and note down the readings. Decrease I_f in steps and note down I_a each time. Plot the V-curves (I_a Vs I_f) for synchronous motor on no-load. Now, for getting V curves of synchronous motor drawing constant power input (say, 800W/phase), vary the lamp/water load and make the wattmeter reading equal to 800W/phase. Increase the field current of the synchronous machine to rated current and the above procedure is repeated. Plot the V curves of the synchronous motor drawing constant power input.

Sl. No.	GENERATING MODE			MOTOR MODE (ON NO LOAD)			MOTOR MODE		
	OUTPUT = _____ W			INPUT = _____ W			INPUT = _____ W		
	I_a	I_f	Cos ϕ (lag/lead)	I_a	I_f	Cos ϕ (lag/lead)	I_a	I_f	Cos ϕ (lag/lead)
1									
2									
3									
4									